

1. O/N/23/22/Q1

Velocity is a vector quantity but speed is a scalar quantity.

- (a) State how a vector quantity differs from a scalar quantity.

.....
..... [1]

- (b) Underline **all** the vector quantities in the list.

acceleration **distance** **energy** **mass** **momentum** **temperature** [1]

- (c) Fig. 1.1 shows a passenger running across a large ship at right angles to the direction in which the ship is moving.

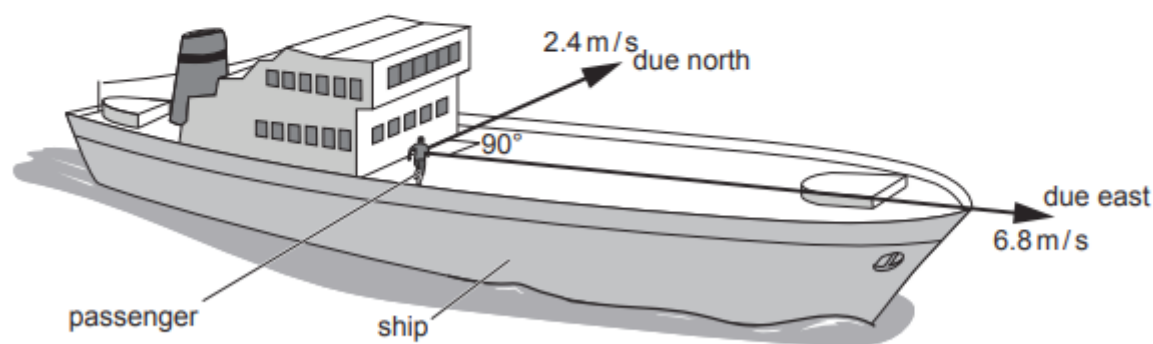


Fig. 1.1 (not to scale)

The passenger runs across the ship at 2.4 m/s due north.

The large ship is travelling due east at 6.8 m/s.

Determine, either graphically or by calculation, the resultant velocity of the passenger. Give the direction in which the passenger is moving relative to due north.

Show how the answer is obtained.

velocity = m/s at ° from due north [4]

[Total: 6]

2. M/J/22/21/Q7

Fig. 7.1 shows a toy helicopter. It can hover and travel through the air.

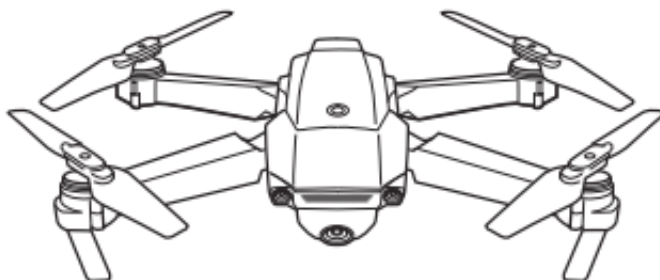


Fig. 7.1

A student flies the toy helicopter on a journey from A to B to C to D at a constant height.

Fig. 7.2 is a scale drawing of the path of the helicopter, viewed from above.

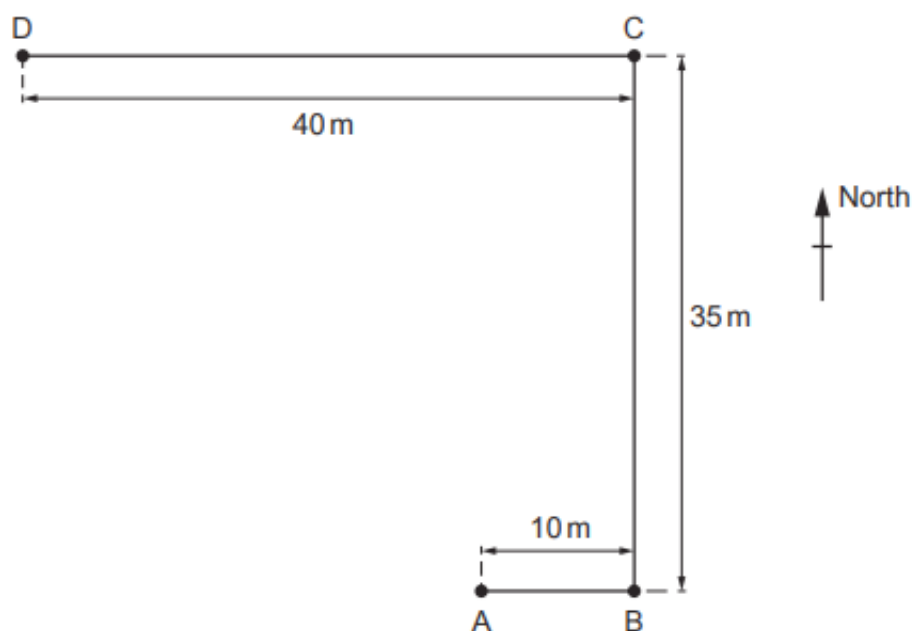


Fig. 7.2 (to scale)

- (a) (i) Determine the total distance travelled by the toy helicopter.

..... [1]

- (ii) The toy helicopter makes the journey in 40 s.

Calculate its average speed during the journey.

speed = [2]

(b) (i) State the difference between distance and displacement.

.....
..... [1]

(ii) Determine the scale used to construct Fig. 7.2 and complete the sentence.

1 cm on the diagram represents on the path. [1]

(iii) Using Fig. 7.2, determine the size of the displacement of point D from point A.

Show your working.

size of displacement = [2]

(iv) Determine the angle between North and the direction of the displacement of point D from point A.

angle = [1]

3. M/J/21/21/Q1

An aircraft flies at a constant height.

Air drag and the force from the aircraft's engines together produce a force on the aircraft of 36 kN due north, as shown in Fig. 1.1.

The wind produces a force of 12 kN towards the east.

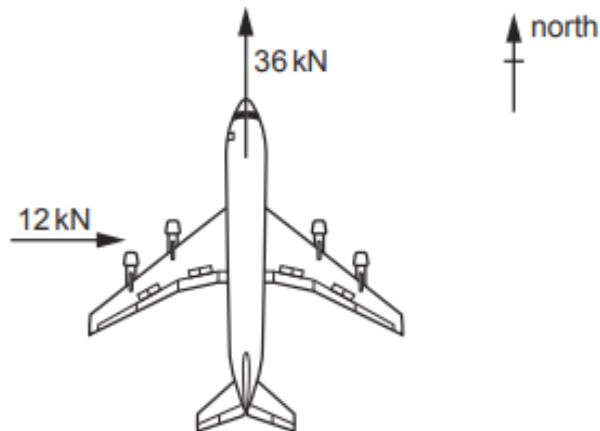


Fig. 1.1 (not to scale)

- (a) Draw a scale drawing to show the resultant force acting on the aircraft.

Use your drawing to determine the size of the resultant force and the angle between the resultant force and north.

size of resultant force =

angle =

[3]

4. M/J/21/22/Q1

(a) (i) State the difference between a scalar quantity and a vector quantity.

.....

.....

..... [1]

(ii) State **two** examples of each type of quantity.

scalar quantity	vector quantity
1	1
2	2

[2]

(b) Fig. 1.1 shows the direction and size of two vectors P and Q.

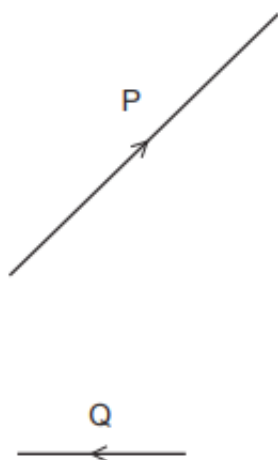


Fig. 1.1

In the space next to Fig. 1.1, draw a labelled vector diagram to show the resultant vector obtained by adding vector P to vector Q.

Draw vector P, vector Q and the resultant vector to the same scale as in Fig. 1.1.

[2]

[Total: 5]

5. O/N/15/21/Q9

Force is a vector quantity and mass is a scalar quantity.

- (a) (i)** State how a vector quantity differs from a scalar quantity.

.....
..... [1]

- (ii)** State one other vector quantity and one other scalar quantity.

vector quantity

scalar quantity [2]

- (b)** An aeroplane flies through the air in a straight line, with its engines at full power. The weight of the aeroplane is 160 kN. There is also a force of 320 kN acting upwards on the aeroplane at 30° to the horizontal.

In the space below, use a graphical method to determine the size and the direction of the resultant of these two forces. Use a scale of 1.0 cm : 40 kN.

size of resultant force =

direction of resultant force =

[4]

6. O/N/13/22/Q1

A set of traffic lights hangs from the end of a metal cable. A horizontal chain pulls the traffic lights to the right so that they are above the middle of the road. Fig. 1.1 shows the metal cable inclined to the vertical.

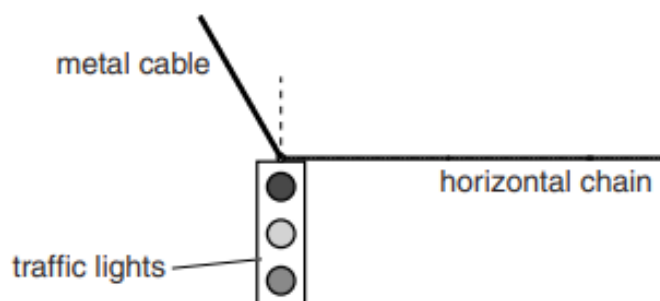


Fig. 1.1

The weight of the traffic lights is 240 N.

- (a) Two of the forces on the traffic lights are the tension in the horizontal chain and the weight of the traffic lights.

On Fig. 1.1, mark

- (i) an arrow that represents the tension in the horizontal chain, [1]
- (ii) an arrow that represents the weight of the traffic lights. [1]
- (b) The tension in the horizontal chain is 140 N. Use a scale diagram to determine the size of the resultant of the weight and the tension in the chain. State the scale used for the diagram.

scale =

resultant force =

[3]

7. O/N/12/21/Q1

A parachutist of total weight 950 N falls vertically at a constant velocity of 6.5 m/s.

- (a)** State the size and direction of the resistive (drag) force acting on the parachutist.

size of resistive force:

direction of resistive force:

[2]

- (b)** At a certain height, the wind starts to blow and the parachutist moves horizontally at a velocity of 4.5 m/s as he continues to fall.

Use a graphical method to add together the horizontal and the vertical velocities to determine the size of the resultant velocity of the parachutist. Label the velocities and state the scale that you use.

size of velocity =

scale =

[3]

8. M/J/10/22/Q1

A journey consists of two displacements: the first is 500m in a northerly direction and the second is 200m in an easterly direction.

(a) In the space below draw, to scale, a vector diagram of these displacements.

State the scale of your diagram.

On your diagram, show the two displacements and the resultant displacement.

Determine the size (magnitude) and direction of the resultant displacement.



scale =

size =

direction =

[3]

(b) Another journey covers a distance of 700m. Describe how it is possible that this journey has no resultant displacement.

.....

.....[1]

9. O/N/08/Q1

A microphone has a weight W of 6.0 N. It is suspended by wire X from the ceiling in a radio studio.

Fig. 1.1 shows the microphone held in the correct position by a horizontal wire Y.

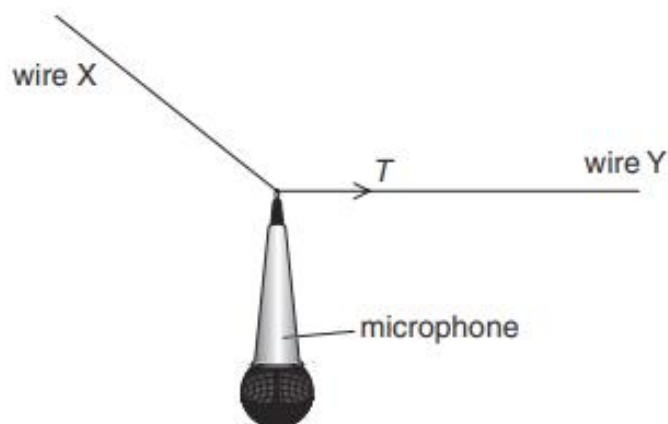


Fig. 1.1

- (a) The tension T in wire Y is 8.0 N. Use a vector diagram to determine the magnitude and the direction of the resultant of W and T .

magnitude =

direction =

[4]

- (b) The microphone is at rest. State the tension in wire X.

tension in X = [1]